DEWDRIVE · RISK REGISTER

DewDrive risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DWD-RSK-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/09-risk.md @ 4f86e9e		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	H4, H6, F5, F11 after Amish's decisions of 2026-10-25 (fan cut-out 65 % RH; 25 m/s design gust)
0.1	2026-10-25	Amish Chadha	First register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 pipeline, stage P3)

DewDrive risk register and hazard analysis

The top project risk is that the design gives too little water: on paper it already misses its 0.5 L per day target (0.44 L), and the real composite may give less than the bulk-salt model. The top hazards are drinking untreated water, burns from the hot interior and $CaCl_2$ in the eyes. The highest design FMEA items are seal leakage at the flaps and lid and damage to the PTFE coating on the brine-wetted parts. Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high); for detection, 5 means hard to detect.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Yield below 0.5 L per day	Already 0.44 L on paper; isotherm and kinetics of the real composite, absorber absorptance, mesh open share and the temperature drop through the bed could take it lower (0.37 L in C13)	PR-001 not met; weaker case for users	5	4	Design lead	Decision requested (DWD-DEC-001, open item 2); measure isotherms and absorptance before building (TRL 4 plan); study the low-emissivity screens with a thermostatic vent (C10)
P2	Cost stays over the value-engineering target	Coatings, glazing grade and the parts that made it buildable	$695 against $520	5	2	Design lead	Savings in DWD-DEC-001 Value engineering; quotes for the estimated lines
P3	No partner site agreed	The field station candidate is a type, not an agreement	No site data, no TRL 4 test	3	4	Amish	Approach two or three stations in parallel (DWD-PLN-001)
P4	Coating or mesh not available locally	PTFE coating shops and PTFE glass-fibre mesh are not everywhere	Brine-wetted parts revert to bare metal and corrode	3	3	Builder	Confirm items 9 and 10 of DWD-DEC-001 before building; fallback is 316 stainless with a shorter life, as a documented decision
P5	Concept choice reopened	A and B within 10 % on the weighted score (DWD-OPT-001)	Rework if Amish chooses B or C	2	4	Amish	Decision requested in DWD-DDR-004
P6	Water quality claims misread	Users treat the water as safe without treatment	Illness	2	5	Partner NGO	Every document says test and treat; labelling on the jerrycan

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Untreated or contaminated water	Dust, microbes or salt in the condensate; jerrycan left for days	Gastrointestinal illness	4	3	Food-grade wetted parts; drip screens and sumps keep brine out; instruction to boil or disinfect; laboratory test at TRL 4	Medium until tested
H2	Hot surfaces inside the box	Bed about 105 °C at noon, 133 °C dry on a peak day (DWD-CAL-001 C2, D6)	Burns to hands	3	3	Open the lid only when cool; gloves; glazing labelled hot; outer surfaces 55 °C or less (PR-013)	Low
H3	$CaCl_2$ contact	Preparing the sorbent; emptying sumps; a split tray	Skin and serious eye irritation (H315, H319)	3	3	Gloves and eye protection (build plan safety stops); pores fill only above 71 % RH; sumps	Low
H4	Brine in the drinking water	Pores overfill in a wet spell; tray tipped; coating failure	Salty water; chloride above 250 mg/L	3	2	Fan cut-out above 65 % RH, restart below 60 % (DWD-DEC-001 item 3); drip screens with no line of sight; level trays; flaps shut in rain	Low
H5	Battery fire or short	Damaged LiFePO4 cell; short in the wiring	Fire, burns	4	1	Built-in BMS, fuse at the battery, shaded IP65 box, LiFePO4 chemistry	Low
H6	Unit tips over in wind	495 N on the lid in the 25 m/s design gust; anchors not fitted	Crush injury, damage	4	2	Two ground anchors rated 250 N or more (or 96 kg of ballast) before the box goes on; check at set-up	Low
H7	Lifting injury	Box 36.1 kg with everything inside	Back injury	3	2	Empty before lifting (label): 23.4 kg, two people	Low
H8	Cuts	Sheet-aluminium edges, fins at ankle and hand height	Cuts	2	3	Deburr all edges; edge breaks in the model; paint on the fins	Low
H9	Glazing failure	Polycarbonate inner skin near 106 °C on a peak day, or UV ageing	Sheet sags or cracks, hot interior exposed	3	2	Sheet rated 130 °C or more (decided 2026-10-02); UV-stabilized	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Inlet flap, lid and outlet flap seals	Keep vapour in by day, force air through the beds at night	Leak	Day vapour lost; night air bypasses the beds; yield falls	EPDM ageing, warped flap, latch not closed	4	4	3	48	Smoke check in the first checks; inspect seals monthly; latches with positive over-centre
F2	PTFE coating on trays, floor bars, absorbers and screens	Keep brine off the aluminium	Coating scratched or pin-holed	Pitting, brine leak, aluminium in the water path	Handling, beads abrading the coating	4	3	4	48	Coating thickness from the coater; handle trays with gloves; inspect at each recharge
F3	Absorber plates	Absorb the sun over the beds	Dust or salt crust on the plate; plate lifted off the bed	Lower absorptance or poor contact; yield falls	Dust through the inlet; brine creep	3	3	3	27	Wipe at recharge; finger holes to lift and reseat; measure absorptance at TRL 4
F4	Composite sorbent	Store water overnight	Capacity loss; gel cracking	Yield falls over the season	Ordinary gel cracking in solution; salt migration	4	3	3	36	Water-resistant gel grade (DWD-DEC-001 item 1); cycling test at TRL 4
F5	Fan and humidity cut-out	Move night air; stop above 65 % RH	Fan stops, or runs in a humid night	No uptake, or pores fill faster	Fan failure; RH sensor fault	3	2	2	12	Logger flags fan current and RH; fail-safe is fan off; sumps as backstop
F6	Battery and charge controller	Power fan and logger	Battery flat	Fan does not run; no log	Cloudy spell longer than 2.6 days; BMS cut-off	2	2	2	8	Logger records battery voltage; 1.48 PV margin
F7	Condenser plate, fins and underside paint	Reject 292 W at an 11.9 K rise	Paint chalks or dust coats the fins; coating on wetted face wears	Condenser hotter; yield falls; aluminium in the water	Sun and dust under the box; abrasion	3	3	3	27	Repaint as needed; log the condenser rise at TRL 4 (PR-005 at risk); food-safe coating check at TRL 4 water test
F8	Gutter and drain	Carry water to the jerrycan	Blocked or leaking	Water lost	Dust, insects, kinked tube	2	3	2	12	Screened drain; tube clip; daily weighing shows a drop
F9	Drip screens and sumps	Catch any brine	Sump overflows	Brine reaches the plate	Wet spell with the	4	1	4	16	Sumps 0.58 L; empty after wet spells

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
					fan rule failed					
F10	Glazing	Let the sun in, hold the heat	Sags or crazes	Heat loss, leak	Sheet below rating, UV	3	2	2	12	130 °C grade; UV-stabilized face up
F11	Stand and anchors	Hold the unit in wind	Anchor pulls out	Tip-over	Soft or sandy soil; gust above the 25 m/s design gust	4	2	3	24	Anchors rated 250 N or more in the site soil (D20; to confirm, item 6); ballast alternative (96 kg); design wind a 25 m/s gust until the site value is known (item 4, decided 2026-10-25)
F12	Tray deck	Carry the trays level on the gasket	Sags past the gasket	Air bypasses at the tray lips	Load at 105 °C	3	2	3	18	Check sag at build (item 8); deeper bar if needed